

Quick Start Guide — Command Station

Get a DCC command station running in minutes

TI MSPM0G3507 LaunchPad + Code Composer Studio / Theia edition. OpenDccCLib itself runs on any microcontroller with a C compiler; this guide covers one concrete path.

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1. What is DCC?

DCC (Digital Command Control) is the NMRA standard for controlling model railroad locomotives and accessories. A command station generates the DCC signal, a modulated square wave on the rails that carries addressed speed, function and programming commands to decoders in locomotives and accessories.

OpenDccCLib is a portable C library that implements the protocol: bit encoding, packet scheduling, service-mode programming and RailCom, so you can concentrate on what your command station does.

1.1 Key Terms

- **Command station:** generates DCC packets on the track.
- **Decoder:** receives and acts on DCC commands, in a locomotive or an accessory.
- **Packet:** one command, address plus instruction plus error check, framed by a preamble and an end bit.
- **Service mode:** the programming-track mode for reading and writing decoder Configuration Variables (CVs).
- **RailCom:** the bidirectional channel in which decoders answer during a short cutout after each packet.
- **Scheduler:** the library component that manages concurrent packets, priorities, repeats and auto-refresh.

2. What You Need

2.1 Hardware

- TI MSPM0G3507 LaunchPad (LP-MSPM0G3507)
- An H-bridge driver board (L298N, IBT-2 or similar) and a 12–18 V supply for track power
- A DCC locomotive with a decoder, or a second LaunchPad running the decoder example
- A USB cable, for programming and the UART terminal

2.2 Signal Pins

All signal pins are on port B; they come from the project's SysConfig file.

LaunchPad pin	Function
PB1	Main-track DCC signal out (DCC_SIGNAL); to the H-bridge input, or to decoder PB1
PB4	Service-track DCC signal out; to decoder PB4
PB12	ACK current-sense in; from decoder PB12
PB17	Track-select out; to decoder PB17
PB2	DCC mirror out (scope aid)
PB3	Debug pulse (scope aid)
PA15	ISR timing pulse (scope aid)
PB22, PB26, PB27	User LEDs; PB22 is the heartbeat
PA10 / PA11	UART TX / RX to the on-board debug probe, 230400 8N1

Two-board setup: CS PB1 to decoder PB1, CS PB4 to decoder PB4, decoder PB12 to CS PB12, CS PB17 to decoder PB17, and GND to GND.

2.3 Software

- TI Code Composer Studio or CCS Theia, with SysConfig (bundled)
- A serial terminal at **230400 baud**, 8N1
- The OpenDccCLib source tree

3. Project Setup

The ready-to-run example is at `applications/ti_theia/mspm03507_launchpad/command_station/`.

1. In CCS, choose File > Import > CCS Projects and point it at that folder.
2. The project reaches the library through the `dcc_lib` symlink to `src/dcc/`; nothing is copied.
3. Build once so SysConfig generates `ti_msp_dl_config.c/h` into `Debug/`.

```
command_station/  
  command_station.c          <- main entry: config struct, ISRs, main loop  
  dcc_user_config.h          <- feature flags and sizes  
  uart_command_parser.c/h    <- UART command-line interface  
  application_callbacks/  
    callbacks_dcc.c/h        <- your application logic  
  application_drivers/  
    ti_driverlib_dcc_driver.c/h <- timers, GPIO, current sense  
    ti_driverlib_uart_driver.c/h <- UART I/O  
  dcc_lib/                   <- the library (symlink; do not edit)  
  Debug/ti_msp_dl_config.c/h <- SysConfig-generated init
```

4. Understanding dcc_user_config.h

This file tells the library which role to compile and how much memory to reserve. Every constant is mandatory and is checked at compile time.

```
#define DCC_COMPILE_COMMAND_STATION
#define DCC_COMPILE_RAILCOM           // cutout + receive
#define DCC_COMPILE_SERVICE_MODE_DIRECT // and PAGED, REGISTER, ADDRESS
#define DCC_COMPILE_SERVICE_MODE_TASK_DIRECT // ... plus the TASK_* orchestrators and TASK_DETECT

#define USER_DEFINED_DCC_SCHEDULER_SLOT_COUNT 24 // concurrent packets
#define USER_DEFINED_DCC_PREAMBLE_BITS_OPS    18 // >= 16 with RailCom
#define USER_DEFINED_DCC_MAX_LOCOS            10 // demo loco table
#define USER_DEFINED_DCC_RAILCOM_BUFFER_DEPTH 4
#define USER_DEFINED_DCC_SERVICE_MODE_RETRIES 3
#define USER_DEFINED_DCC_ACK_THRESHOLD_MA     60
#define USER_DEFINED_DCC_ACK_MIN_DURATION_US  5000
#define USER_DEFINED_DCC_ACK_MAX_DURATION_US  7000
#define USER_DEFINED_DCC_ACK_DROPOUT_TOLERANCE_US 116
```

The service-mode flags require DCC_COMPILE_COMMAND_STATION; the compiler stops with a #error otherwise.

5. Building and Flashing

1. Build the project (Project > Build, or Ctrl+B).
2. Connect the LaunchPad over USB.
3. Flash and run (Run > Debug, or F11).
4. Open a terminal at 230400 baud on the LaunchPad's serial port.
5. The banner reads DCC Command Station - MSPM0G3507 LaunchPad.

5.1 What Happens at Power-On

`main()` initializes the drivers, hands the `dcc_config_t` struct to `DccConfig_initialize()`, then loops. Track power is off until you type `POWER ON`.

```
while (1) {  
    DccConfig_run();           // scheduling, callbacks  
    TI_UartDriver_echo_process(); // echo typed characters  
    UartCommandParser_process(); // parse and execute commands  
}
```

6. Using the UART Command Interface

Type HELP for the full list. Every command answers OK: ... or ERR:

Command	Description			
`POWER ON	OFF`	Track power		
`SPEED <addr> <speed> <FWD	REV> [14	28	128]`	Speed and direction; auto-refreshed (sent 3 times at once, then kept alive about every 0.4 s)
ESTOP [addr]	Emergency stop, one loco or broadcast			
STOP	Broadcast controlled stop			
`FUNC <addr> <0-68> <ON	OFF>`	Function on or off; auto-refreshed the same way		
`ACC <board> <pair> <ON	OFF> / ACCE <addr> <aspect> / NOP <addr> [E]`	Basic accessory, extended accessory, accessory NOP		
ACC CV ... / ACCE CV ...	Accessory operations-mode CV write, verify, bit			
`CV WRITE	VERIFY <addr> <cv> <value> / CV BIT <addr> <cv> <bit> <0	1>`	Loco operati ons-mode CV access	
`CONSIST <addr> SET <ca> [NORMAL	REVERSE] / CONSIST <addr> CLEAR`	Advanced consist		
BSS, BSL, ANALOG	Binary state short and long, analog function			
SYSTIME <ms> / MTIME ... / MDATE <d> <m> <y>	Broadcast time and date			
SVC ENTER / SVC EXIT / SVC DETECT	Service mode on the programming track			
`SVC DIRECT	PAGED	REG	ADDR ...`	Read, write and verify CVs in each mode
`REFRESH ON	OFF / CLEAR / RESET`	Auto-refresh policy, clear the scheduler, broadcast reset		

Command	Description
STATUS / HELP	Status line, command list

6.1 Quick Test

```
> POWER ON
OK: track power ON
> SPEED 3 50 FWD 128
OK: SPEED addr=3 speed=50 dir=FWD mode=128
> FUNC 3 0 ON
OK: FUNC addr=3 F0=ON
> ESTOP
OK: ESTOP broadcast
> POWER OFF
OK: track power OFF
```

A one-shot command such as CV WRITE goes out the number of times the standard requires (twice for a CV write); the library's packet builders carry those counts.

7. What's Next

- **RailCom detection.** Wire a RailCom detector's UART output to a 250 kbaud UART, fill a `dcc_railcom_hw_t` and point `main_track.railcom` at it. The library runs the cutout, gates the receiver to the reply windows and delivers decoded datagrams to `on_railcom_datagram_result`. The receive hooks in `dcc_config.h` state the gating contract.
- **Current-sense protection.** Set `main_track.current_sense_read` to an ADC or comparator read.
- **Another MCU.** Copy the project and rewrite the two driver files for your timers and GPIO. The library never touches hardware directly.
- **Developer Guide.** The companion Developer Guide, Command Station, explains the config struct, scheduler, bit encoder, service mode and RailCom in depth, and the hardware-in-the-loop bench under `test/compliance/` that proves the wire.